



POLITECNICO MILANO 1863

Requirements Analysis Specification Document

Ali Noshad, ali.noshad@mail.polimi.it, 101108

November 15, 2024

Students&Companies - Requirements Analysis Specification
Document

Prof. Matteo Rossi

Contents

1	Introduction	4
1.1	Purpose	4
1.1.1	Goals	5
1.2	Scope	5
1.2.1	World Phenomena	6
1.2.2	Shared Phenomena	7
1.3	Definitions, Acronyms, Abbreviations	7
1.3.1	Definitions	7
1.3.2	Acronyms	7
1.3.3	Abbreviations	7
1.4	Revision history	8
1.5	Reference Documents	8
1.6	Document Structure	8
2	Overall Description	8
2.1	Product perspective	8
2.1.1	Scenarios	8
2.1.2	Class Diagram	10
2.1.3	State Diagram	10
2.2	Product Functions	18
2.3	User Characteristics	18
2.3.1	Actors	18
2.4	Assumptions, Dependencies and Constraints	18
2.4.1	Domain Assumptions	18
2.4.2	Dependencies	18
2.4.3	Constraints	18
3	Specific Requirements	18
3.1	External Interface Requirements	18
3.1.1	User Interfaces	18
3.1.2	Hardware Interfaces	18
3.1.3	Software Interfaces	18
3.1.4	Communication Interfaces	18
3.2	Functional Requirements	18
3.2.1	Use Case Diagram	18
3.2.2	Use Cases	18
3.2.3	Sequence Diagram	18
3.2.4	Goal Mapping on Requirements	18
3.3	Performance Requirements	18
3.4	Design Constraints	18
3.4.1	Standard Compliance	18
3.4.2	Hardware Limitations	18
3.4.3	Any Other Constraint	18
3.5	Software System Attributes	18
3.5.1	Reliability	18
3.5.2	Availability	18
3.5.3	Security	18
3.5.4	Maintainability	18
3.5.5	Portability	18
4	Formal Analysis using Alloy	18
4.1	What we want to model?	18
4.2	Alloy Model	18
4.3	Alloy Generated Worlds	18

1 Introduction

The Requirement Analysis and Specification Document (RASD) aims to focus on the tasks needed to develop and implement an application, taking account of the requirements of the involved stakeholders and, analyzing and documenting also the application requirements. Then, in the second part of the document, there is a more formal definition of the requirements with the use of the Alloy language. In general this document is meant for developers tasked with the implementation of the system and also for all the other entities involved in validation and managing of the project.

1.1 Purpose

Recently job market becoming more and more competitive. Accordingly students face significant challenges in finding suitable job opportunities and internships with respect to their experiences, skills and most importantly their interest and their career goals. Simultaneously, companies find it hard who with fit their need and requirements as well as their work culture. The Students&Companies (S&C) platform addresses this need by streamlining and enhancing the process of matching students with relevant internship opportunities. By analyzing the experiences, skills and attitudes of students listed in their CVs and comparing them with the specific requirements and perks of available roles provided by the companies, S&C connects students with positions that suit them well.

The platform helps students easily find internship opportunities by collecting detailed information about available internships, student resumes, skills, and preferences. This allows students to search for companies and roles that match their interests through S&C's search system, giving them more control over finding an internship. Additionally, S&C assists companies by managing interview schedules and assessments for students who apply for specific roles, making the hiring process smoother and helping both sides make better decisions. Finally, S&C offers tools to track internships by gathering student feedback and complaints, providing insights to improve future programs. Overall, S&C aims to connect students and companies, making it easier and more transparent for students to start their careers.

This document focuses on the Requirements Analysis and Specification Document (RASD) of the system and describes the main goals, the domain assumptions, the scenarios which may happen, the uses cases, the list of functional and non-functional requirements which system should fulfill, and finally the diagrams to visualize the interactions between components and performance of the system.

1.1.1 Goals

Table 1: Goals Definition

Goals	Description
G1	S&C allows students to manage (create, upload, edit) their resume
G2	S&C allows students to see available internships (e.g., by location, industry)
G3	S&C allows students to initiate the search process and submit applications directly
G4	S&C allows students to track the status (reviewed, shortlisted, selected) of their internship applications
G5	S&C allows students to raise complaints or concerns regarding an internship experience
G6	S&C allows companies to manage (create, edit, publish, delete) internship listings
G7	S&C allows companies to browse and search through student profiles (CVs) applied for the internship (e.g., by location, industry)
G8	S&C allows companies to review incoming applications
G9	S&C allows companies the ability to shortlist candidates
G10	S&C allows companies communicate with applicants
G11	S&C allows companies to provide updates on their application status of incoming applications
G12	S&C allows companies to manage (create, edit, send) questionnaire to applicants
G13	S&C allows companies to manage (scheduling and conducting) interviews with applicants
G14	S&C allows universities to track application processes
G15	S&C allows universities to monitor internship execution
G16	S&C allows universities to collect complaints

1.2 Scope

The Students & Companies (S&C) platform connects and facilitate matching university students want to find internships with companies providing those opportunities. The platform will provide abilities for students to manage their profiles, apply for internships, and track their application status. At the same time, it will allow companies to post internships offerings, review applications, and communicate with potential candidates. The platform also includes an access for universities to monitor internship activities and collect compliance.

To initiate application process, an application has been defined through which the user can do the following:

Student Features

- Students must register to system with their credentials.

- Student must log in to his/her account. Then, students can create, upload, and edit their resumes.
- Then, students can view available internships and filter them based on various criteria.
- After selecting desired internships, students can initiate applications for internships and track the status of their applications (e.g., reviewed, shortlisted, selected).
- Furthermore, students can raise complaints or concerns regarding their internship experiences.

Company Features

- Companies must register to system with their credentials.
- Companies must log in to their account, and then, an create, edit, publish, and delete internship listings with all the necessary information (e.g. requirements, details of the job, ...)
- Then, for each internship posting, companies can browse and search through student profiles (CVs) that have applied for their internships.
- After receiving applications, companies can review incoming applications and shortlist candidates based on their qualifications.
- Then, companies can communicate with applicants and provide updates on application statuses.
- Companies can schedule and conduct interviews with shortlisted candidates and manage questionnaires for applicants.
- Finally, companies can decide which applicants are suitable for the positions based on interview and questionnaire.

Universities Features

- universities must register and login to their account.
- Then, universities can monitor the application processes of their students.
- Also, universities can oversee the execution of internships to ensure compliance and quality.

1.2.1 World Phenomena

Table 2: World Phenomena

World Phenomena	Description
WP1	Students actively look for internship opportunities based on their interest and skills and knowledge
WP2	Students prepare CVs
WP3	Companies offer various internships with specific requirements, expectations, and work cultures, which are of interest to potential student applicants.
WP4	Companies review students CVs
WP5	Companies decision on short listing candidates
WP6	Companies decision on acceptance/rejection of candidates
WP7	Students accept/reject the offer of internship
WP8	Students start the internship

1.2.2 Shared Phenomena

Table 3: Shared Phenomena

Shared phenomena	Description	Control
SP1	Students create, upload, and edit their resumes within the system to apply for internships	World
SP2	Companies create, publish, edit, and delete internship listings through the platform	World
SP3	System shows the available internships	System
SP4	Students use the system's search and filtering functions to explore available internship opportunities	World
SP5	Students apply to the internship through the system	World
SP6	Companies search and filter applicants through the system	World
SP7	Companies manage (create, edit, publish, delete) questionnaire through the system	World
SP8	Companies schedule a meeting through the system	World
SP9	Students are notified about a change in applications	System
SP10	System shows all the students applied for an internship	System
SP11	System notifies the universities about the status and process of internship	System
SP12	System notifies the companies about whether the internship accepted/rejected	System

1.3 Definitions, Acronyms, Abbreviations

1.3.1 Definitions

1.3.2 Acronyms

1.3.3 Abbreviations

Table 4: Abbreviations

Abbreviations	Description
G	Goal
R	Requirement
C	Component
WP	World Phenomena
SP	Shared Phenomena

1.4 Revision history

1.5 Reference Documents

- Specification Document: “Assignment RDD AY 2024-2025.pdf”
- Course Slides

1.6 Document Structure

Section 1 Introduces the problem and the purpose of the project and defining the scope of the Students&Companies. Describe the specifications such as the definitions, acronyms, abbreviations, revision history, and references. As well as introducing the goals, world and shared phenomena of the software.

Section 2 Identifying key scenarios and detailing the software’s primary features through class diagrams and state charts. The user characteristics section describes the types of actors interacting with the application, while the product function subsection outlines its functionalities. Finally, domain assumptions are established.

Section 3 The main part of the project which introduces interface requirements such as user interface, hardware interface, software interface, and communication interfaces. Presenting the functional requirements that are shown by use case diagrams and sequence diagrams. Then requirements are mapped to use cases.

Section 4 Using Alloy language for analyzing the system and brief comments for clarifying the Alloy codes.

Section 5 Shows how much time is spent on this project.

Section 6 Contains the references

2 Overall Description

2.1 Product perspective

2.1.1 Scenarios

Student Uploads CV Mia is sitting at her desk at the university library, she is excited to showcase her skills, capabilities, and experience on the S&C platform. She navigates to her profile page, where she finds an option to upload her CV. Mia selects ”Add CV”, which opens a file explorer window. With precision, she locates her CV file within the organized ”Career” folder on her desktop. She clicks to upload the document. The platform start the upload process, automatically checking the document format to ensure compatibility. A progress indicator shows the confirmation of the upload, and once completed, Mia can review her uploaded CV to confirm accuracy and ensure it aligns with her career goals and information.

Company Posts Internship Alex is a representative of company called BigTech and working in hiring manage department. She has been notified by her boss that a new intern is required for R&D department. Accordingly, Alex is preparing to post a new internship. He logs into the platform, selects “Post Internship,” and begins to fill out the details. He describes the responsibilities, the technical skills required, and any perks like mentorship or training programs. After adding all the information, Alex hits “Submit.” Within moments, Mia and other students can now view the new opportunity and consider applying.

Viewing internship opportunities On the S&C platform, Emma uses the search feature to find internships by entering keywords like ”machine learning,” ”AI,” and ”data analysis.” She filters results by location, seeking options close to her university campus, and sorts the list by the latest postings. After browsing several listings, Emma finds the TechVision Corp internship. she clicks in to learn more, reading through the description to see if it aligns with her career aspirations. She reviews the details of the position, compares it with other internships, and decides it’s a good match.

Student Initiates Application After reading through the details, Mia feels Alex’s internship is a perfect match for her goals. She clicks “Apply,” and her CV is immediately sent to Alex’s company profile. The platform confirms her application, and Mia’s dashboard updates to show the internship as “Applied.” She’s thrilled to take her first step towards the role.

Company views all the applicants for internship posting Sarah, an HR manager at TechVision Corp, has posted an internship for a machine learning role on the S&C platform. After several days, she has received numerous applications. Sarah logs into the S&C platform and navigates to the “Internships” dashboard for her internship posting. The platform provides a list of applicants, along with their CVs. Sarah uses the platform’s filtering options to shortlist applicants based on criteria such as relevant skills and project experience, enabling her to focus only on the most qualified candidates.

Scheduling Interviews by company’s hiring manager After reviewing the applications, Sarah decides that Emma and a few other candidates are well-suited for the role. She wants to set up interviews to get to know them better. Sarah selects Emma’s profile and clicks “Schedule Interview” on the S&C platform. A calendar view appears, showing Emma’s availability based on her provided schedule, along with suggested time slots to minimize conflicts. Sarah picks an available slot and sends the interview invitation. Emma receives a notification on the S&C platform, confirming the interview time and including a link to join the online meeting. Sarah picks an available slot and sends the interview invitation. Emma receives a notification on the S&C platform, confirming the interview time and including a link to join the online meeting.

Creating a Questionnaire for Candidate Assessment To streamline the interview process, TechVision Corp uses a structured questionnaire to gauge candidates’ technical knowledge and problem-solving skills. Sarah logs into the S&C platform and navigates to the “Assessment Tools” section. Here, she finds options to create a new questionnaire or select pre-made templates relevant to her internship role. Since she wants to evaluate specific skills related to TechVision’s project requirements, Sarah opts to create a new questionnaire from scratch. She decides that the questionnaire should cover three main areas: technical knowledge (multiple-choice), coding skills (coding challenges), and scenario-based problem-solving (short-answer questions). After building the questionnaire, Sarah reviews each question for clarity, relevance, and alignment with the internship’s objectives. She configures the scoring criteria for each section. After the questionnaire has been completed, she send and publish this questionnaire to all the applicants applied for the internship position.

Answering to questionnaire Emma logs into her S&C account and sees a notification on her dashboard about a new task. She clicks on the alert, which opens a detailed view of her application progress. A message from TechVision Corp indicates that she needs to complete a pre-interview questionnaire within three days. Emma also receives an email reminder with a direct link to the questionnaire. Emma clicks on the link in her dashboard, which takes her to the “Assessment” section of her application profile. Here, she sees an overview of the questionnaire, including the expected completion time. After completing all sections, the platform allows Emma to review her responses. She takes a final look to confirm that she hasn’t missed any questions and feels satisfied with her answers. Emma clicks “Submit,” and the S&C platform displays a confirmation message, assuring her that her responses have been successfully sent to TechVision Corp’s HR team.

Finalizing the Evaluation and Making a Decision After interviewing multiple candidates, Sarah accesses the S&C platform to review her evaluation notes, assessment scores, and feedback from each candidate’s questionnaire. After finalizing her decision, Sarah updates Emma’s application status to “Accepted” on the S&C platform. This change automatically notifies Emma, informing her that she has been selected for the internship and can review her offer details.

University Internship Monitoring The university representative, Professor Green, logs into the S&C platform. Upon accessing the university dashboard, she sees an overview of all ongoing internships involving students from her university. Professor Green selects the “Internship Monitoring” tab, where she views a list of students currently interning, along with detailed internship information such as company names, role titles, start and end dates, and supervisors. Each entry shows the student’s

progress and any recorded evaluations or feedback from the company.

Student Complaint Submission Mia, a student currently in an internship, encounters issues regarding her working hours. She logs into her S&C profile, navigates to "Internship Support," and selects "Submit Complaint.". Mia fills in the complaint form, specifying the issue and selecting "Excessive Work Hours" from a list of predefined categories. She adds further comments to explain her situation and attaches relevant evidence, such as email correspondence with her university supervisor.

Receiving Complaint By University Professor Green, who is responsible for overseeing student internships at the university, wants to check if any students have submitted complaints regarding their internships. Using the S&C platform, she can efficiently access and manage all student complaints in one place. Professor Green logs into the S&C platform and navigates to the "Complaints" section in her dashboard. Here, she has a dedicated view displaying all submitted complaints from students at her university. On this page, she sees a list of complaints categorized by student names, company names, submission dates, and complaint categories (e.g., "Work Hours," "Workplace Environment,"). Each entry also has a brief description of the issue for quick reference. To prioritize her review, Professor Green uses filtering options to sort complaints by severity, date submitted, or status (e.g., "New," "In Progress," "Resolved"). This helps her quickly identify urgent issues.

2.1.2 Class Diagram

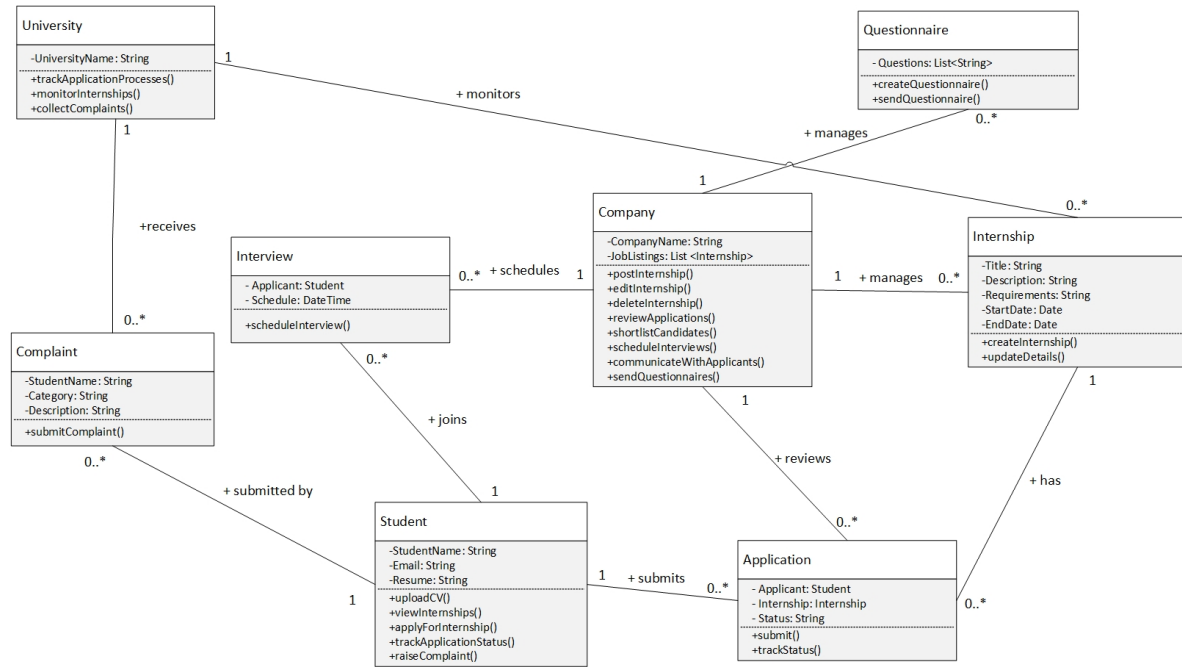


Figure 1: Class Diagram

2.1.3 State Diagram

Login. If an already registered user wants to login into their S&C account, they must provide the login credentials which includes email and password used in the time of the registration. If the credentials are accurate-matching those of the registered user. S&C shows the user's homepage, showcasing Internships, and on going Applications in which the user participating in. On the other hand, if the credentials are invalid up to third try, the login will be denied.

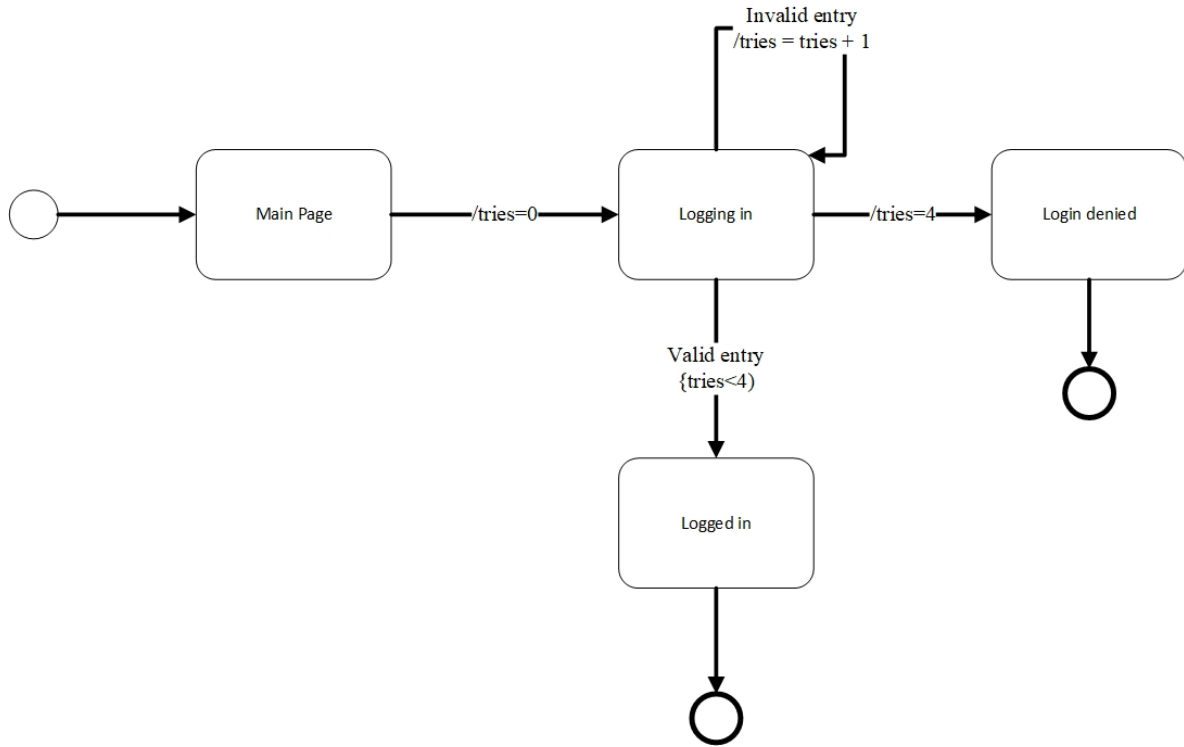


Figure 2: Login State Chart

Posting internships. When and Company User wants to post an internship, he/she is required to fill a form with specific details about the internship in the form, including job title, requirements, location, and date and application deadline or any specific rules associated with this position. If any of the entered parameters/details are known invalid by the system, S&C shows an error message, and redirect the user back to the post internship page. If all the entered parameters/details are meet the system's criteria, then, S&C create the internship posting. The new created internship is then added to the list of internships.

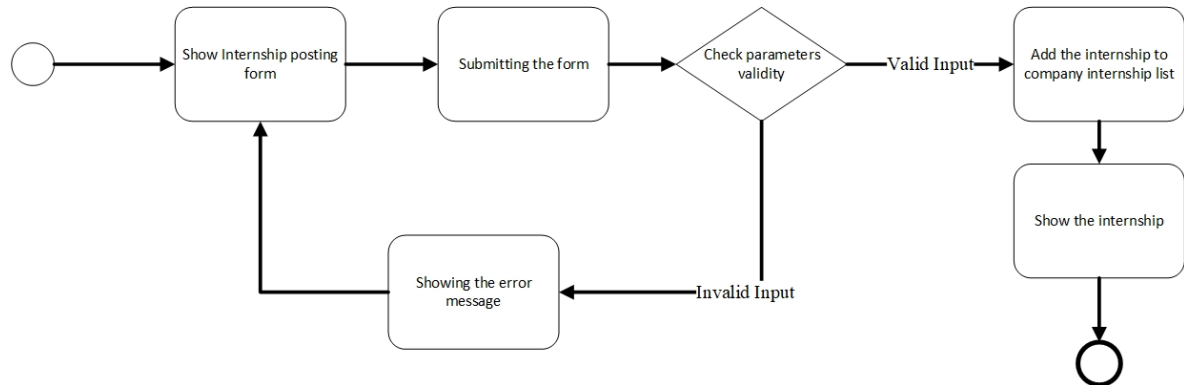


Figure 3: Internship Posting State Chart

View internship listing. With the use of the search bar user can view internships that they are looking for, just by entering the name or a keyword with respect to the details they want to find. When the parameter is entered, S&C will look for the internships that contain the entered parameter. Then the search results are shown to the user. Then user can click on specific internship from the search results, and S&C redirect user to the internship dashboard for detailed information and interaction with the internship.



Figure 4: Search and View Internships State Chart

Send application by a student. Once the student found the specific internship which matches for her/his interest and skills. Student has an option to apply and send his/her resume. S&C first performs a check to valid the application deadline of the internship. If student is applied throughout the allowed time-frame (before the deadline), S&C adds the students resume and details to the list of all the applicants applied for that internship, and directs student to applied internship page in their dashboard. On the other hand, if student is trying the send the application after the application deadline, S&C shows an error message to the student inform them about the closed application portal and expiration, and redirects student to the internship listing page for finding another internship.

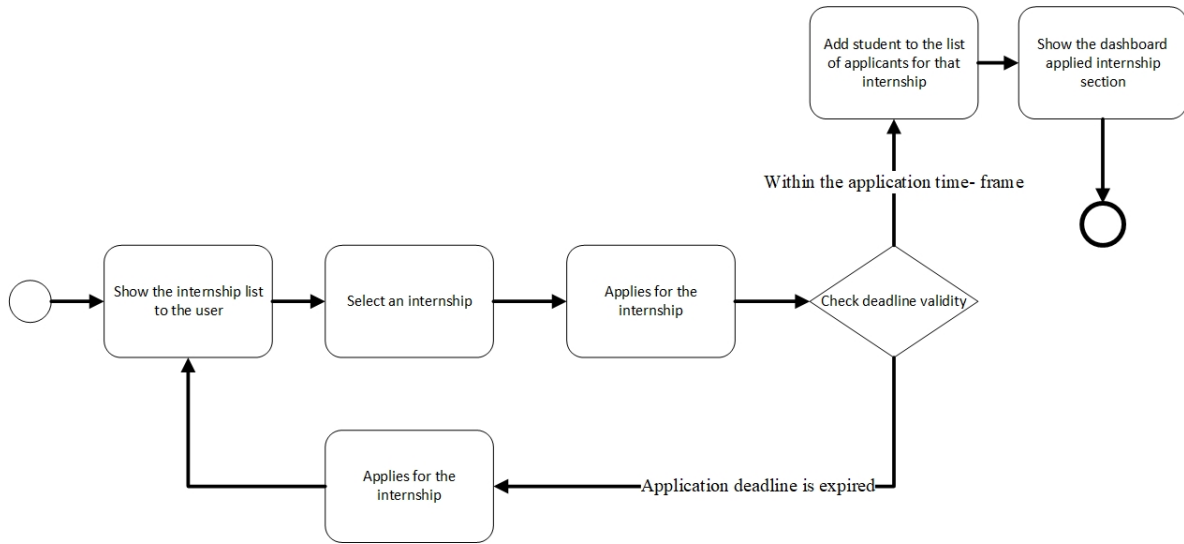


Figure 5: Sending Application for an Internships State Chart

View and shortlist the applicants for an internship by a company. A company user can goes to the internships listing page in the his/her dashboard, the user see all posted internship, and then, by selecting an internship, the user can see all the applicants applied for that position, and by clicking on a specific applicant user will redirect to a page that shows all the information about the applicants including resume. Furthermore, there is a search bar, by which the user can search specific keyword in the applicants entered information. Then, S&C shows the list of applicants matching the search keyword entered by the user. Then, user can short list the applicants after reviewing their qualifications.

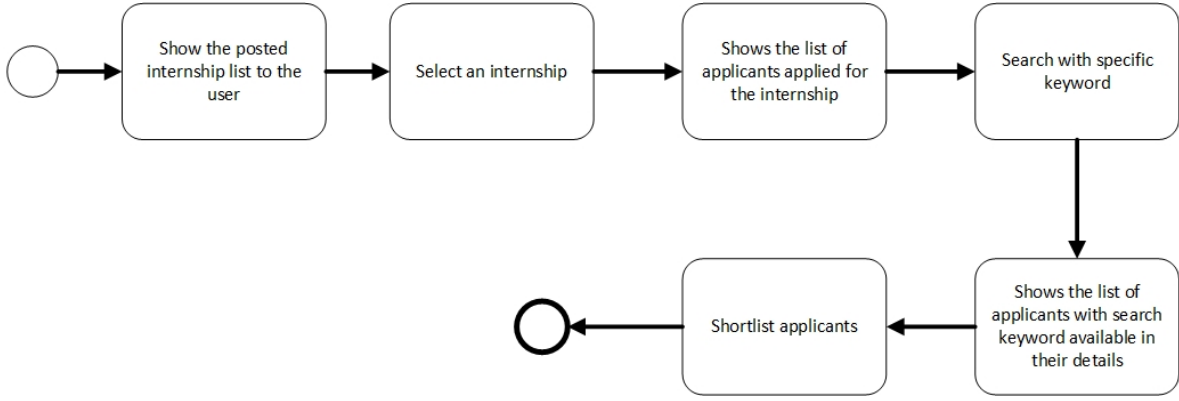


Figure 6: View and Shortlisting Applicants for an Internship State Chart

Scheduling an interview by a company. The process of scheduling an interview begins with a company representative logging into the platform and accessing their dashboard, where they can review the list shortlisted of applicants for a specific internship. After selecting an applicant for an interview, the representative navigates to the scheduling section and chooses a suitable date. Then, the S&C system checks the availability of the applicant in the selected date and time, if the applicant is not available, the S&C system displays an error message informing the company user about the availability, and company user should try with another time. On the other hand, if the applicant is available in that time, S&C system sends a meeting proposal to the applicant, if the applicant is confirmed the interview, S&C system will set the meeting date and time in the applicant's calendar. However, if the applicant does not accept the interview, the applicant will be rejected.

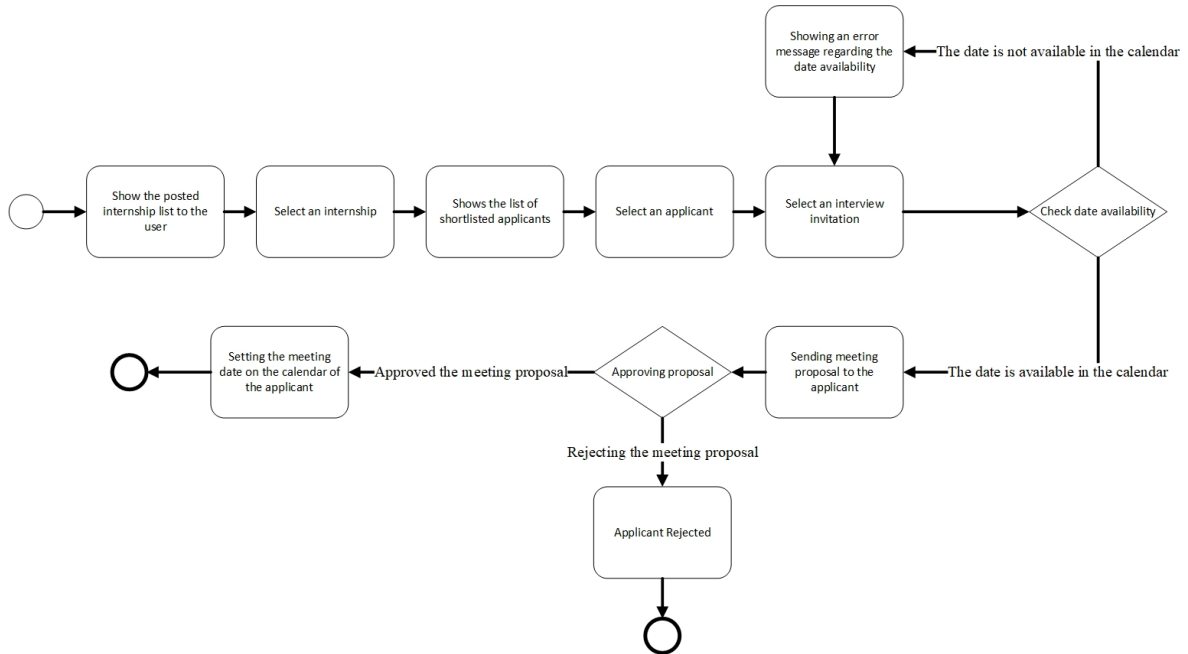


Figure 7: Scheduling Interview by a Company State Chart

Creating a questionnaire by a company. The process of creating a questionnaire for an internship begins by an already logged in company user, goes to section of posted internships, and by selecting an internship, company user navigate to creating a questionnaire form, and complete the form with question and details that the user wants to know about the applicants. After, the completion, the company user submits the form, the S&C system will check that if all the entered inputs are valid with respect to the criteria of the system, if the they were valid, the S&C system adds the questionnaire to the internship posting, and then sends notification to all the candidate that applied for that internship

about questionnaire. However, the inputs are invalid with respect to the previous defined structure in the S&C system, the S&C system will display an error message to the company user, the it redirects the user to the posted internship page.

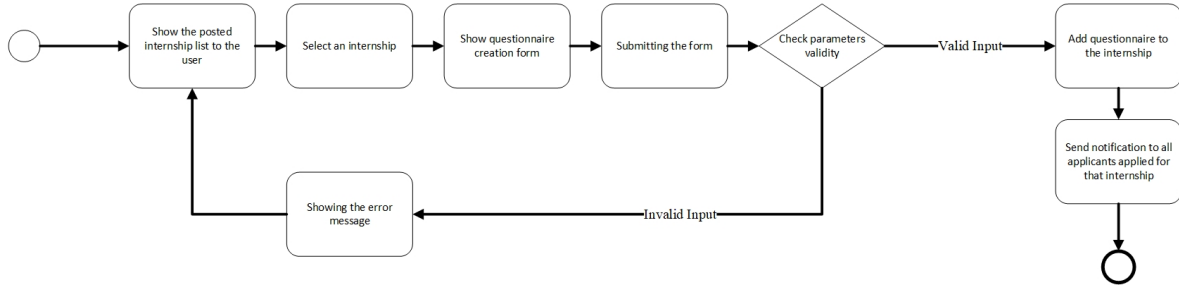


Figure 8: Creating a Questionnaire by a Company State Chart

Answering the questionnaire by a student. After that the company, create a questionnaire for an internship that they posted, the notification is sent to all the applicants applied for that internship. Then, the student sees the notification, and considering that he is already logged in to the S&C platform, the student user goes to the application section, in which the user sees all the applications that is sent for internships. Then, by selecting the the specific internship application, he/she will be go the detail page of the application, in which the user can see the application, and the new questionnaire associated with that application. Then, the user navigates to the option of filling the questionnaire. If this form has already been completed by the user, the S&C system shows an error message to the user and notify the user that the user about the form has already been filled. On the other hand if this is the first time, the user will be redirected to questionnaire page, the user then reviews all the instructions associated with this assessment, and then the user fills the form and submits. Following this, the S&C system will check the validity of the inputs, if all the inputs are valid the questionnaire will be submitted, and the user will be redirected to application page. However, if the input is invalid, the S&C shows the error message to the user.

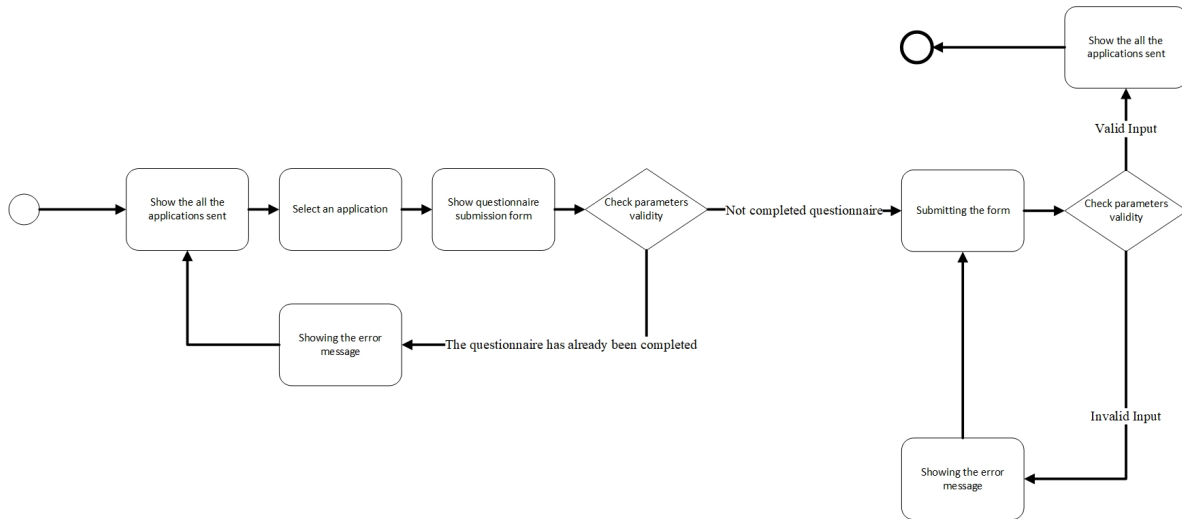


Figure 9: Answering a Questionnaire by a Student State Chart

Making the final decision about an internship by a company. Considering an already logged in company user wants to make final decision about an internship. The user goes on posted internship page, and then the user selects the internship that have a final decision. Then, the user view all the shortlisted applicants for that internship. Then, the user selects an applicant for the offer. First, the S&C system checks the availability of the applicants regarding the internship period. If the applicant is not available or he/she has another internship in progress during that period, the

applicant will be rejected, and the S&C shows an error message informing the company user about the situation, and then S&C redirects the company user to the shortlist page of the applicants. However, if the applicant is available, the company user sends the internship offer to the applicant, and awaits the applicant response. If the applicant accept the offer, the system will automatically change the status of all other applicant to rejected and the applicant of interest to accepted, and finally the system sends notification to all the applicants. However, if the applicant reject the internship offer, the system update the status of applicant and send the notification of rejection to that applicant.

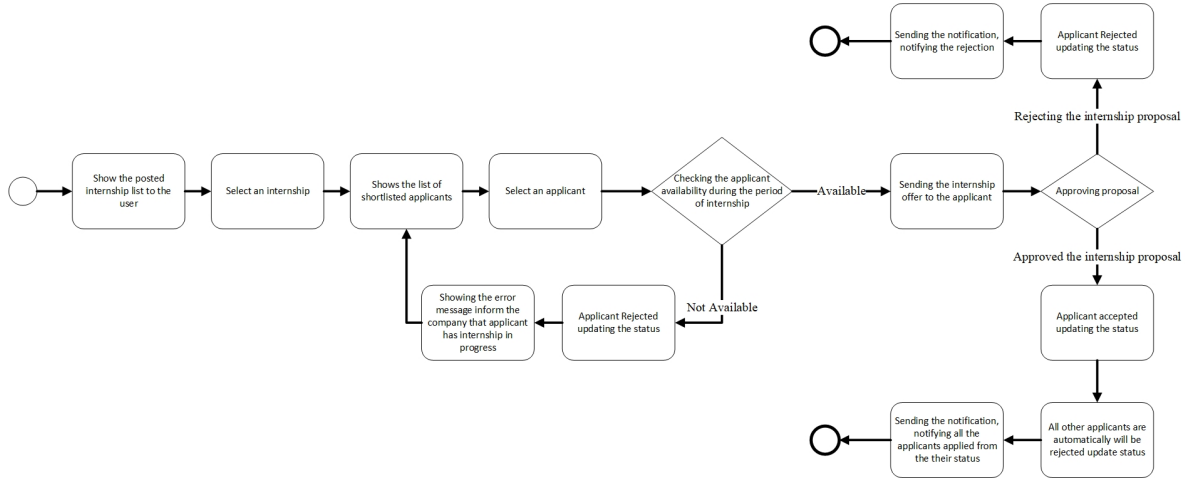


Figure 10: Making a Final Decision by a Company State Chart

University monitors an internship. Considering the university representative user is already registered and logged in to the websites. The user then goes to the on-going internship page, which represents all the internships that involve their students (this is identified during the registration time when the students insert the university affiliation). By selecting the an internship, they can see all the detail about the progress of the student that is taking the internship.



Figure 11: Monitoring Internship by a University State Chart

Student submits an complaint for an internship to university. Considering an already registered and logged in student, with correct affiliation of the university. Student user goes on the page of on-going internship, in which he sees all the internship that user involved such as, applied for, rejected, accepted or in-progress. The student user then selects the internship that the user wants to submits a complaint for. Then, by accessing he complaint form in the internship page, the student clicks on filling the form. The S&C system first check if the student is somehow involves in the internship or not (based on the status that described above). If the student is involved in the internship, he/she can submit the form. Then the S&C checks that the input is valid, if the input is valid, the form will be submitted, and the status of complaint will be set to submitted, and the S&C sends notification to the representative university of the student. However, if the input entered by the student is invalid, the S&C shows an error message to the student, and informing the user about the wrong input, and redirects the student to the list of on-going internship page. Furthermore, if the student is not involved in the internship by all means, the S&C system will return an error.

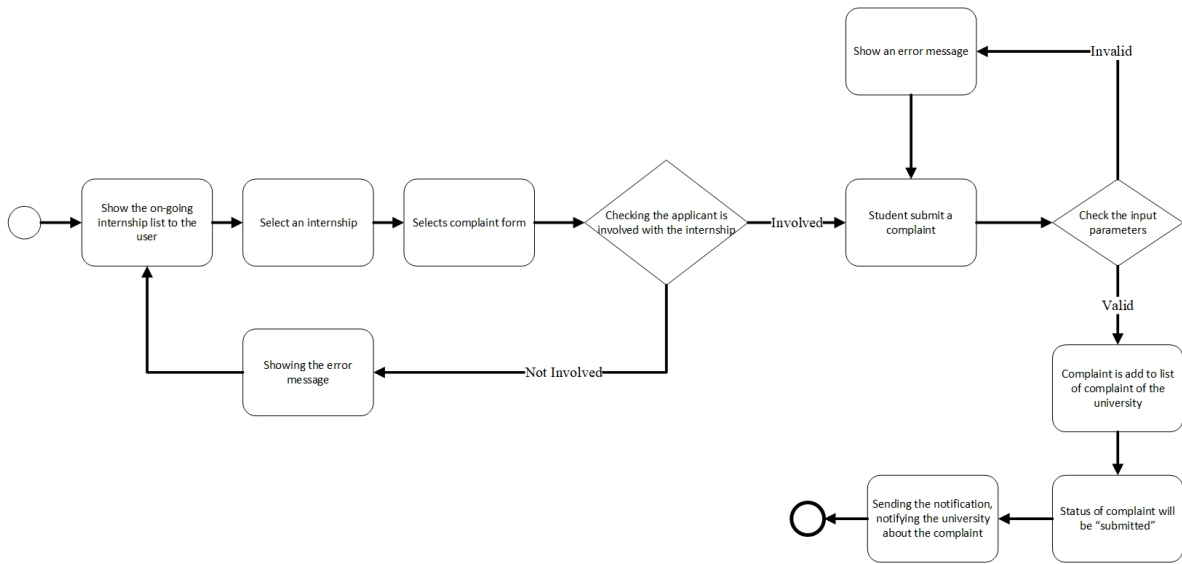


Figure 12: Submitting an Complaint by a Student State Chart

Receiving complaint by the university. Considering the university representative user is already registered and logged in to the websites. The user then goes to the on-going internship page, which represents all the internships that involve their students (this is identified during the registration time when the students insert the university affiliation). By selecting the an internship, they can see all the detail, and all the complaint submitted by their student to that internship. By clicking on the complaints, the user goes to page of complaints of that internship, then, the university user selects an specific complaint. After selecting, the S&C system changes the status of complaint to "In-Progress".

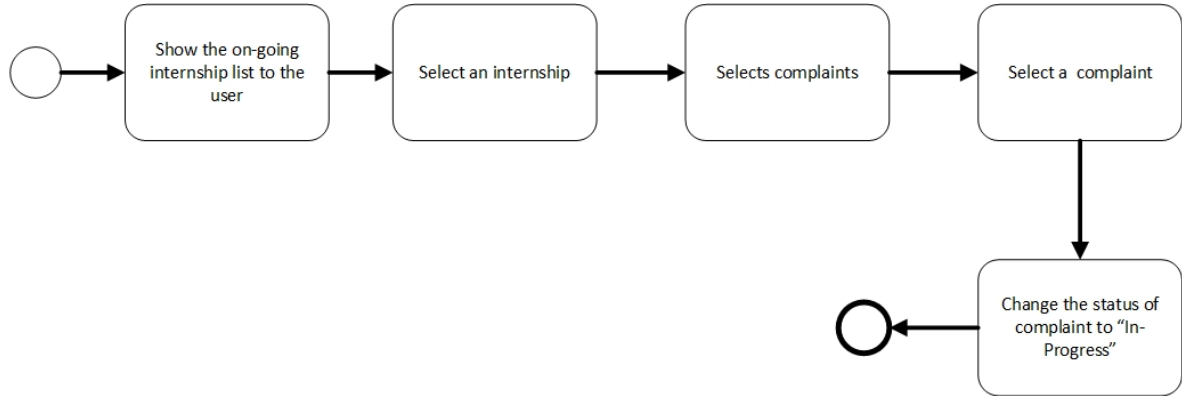


Figure 13: Receiving Complaint by a University State Chart

2.2 Product Functions

2.3 User Characteristics

2.3.1 Actors

2.4 Assumptions, Dependencies and Constraints

2.4.1 Domain Assumptions

2.4.2 Dependencies

2.4.3 Constraints

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

3.1.2 Hardware Interfaces

3.1.3 Software Interfaces

3.1.4 Communication Interfaces

3.2 Functional Requirements

3.2.1 Use Case Diagram

3.2.2 Use Cases

3.2.3 Sequence Diagram

3.2.4 Goal Mapping on Requirements

3.3 Performance Requirements

3.4 Design Constraints

3.4.1 Standard Compliance

3.4.2 Hardware Limitations

3.4.3 Any Other Constraint

3.5 Software System Attributes

3.5.1 Reliability

3.5.2 Availability

3.5.3 Security

3.5.4 Maintainability

3.5.5 Portability

4 Formal Analysis using Alloy

4.1 What we want to model?

4.2 Alloy Model

4.3 Alloy Generated Worlds

5 Effort Spent

References